TE WHARE WĀNANGA O TE ŪPOKO O TE IKA A MĀUI
VICTORIA
UNIVERSITY OF WELLINGTON
SCHOOL OF ECONOMICS
AND FINANCE

ECON412 INTERNATIONAL ECONOMICS: TRADE

TRIMESTER 2, 2018

LECTURE WEEK #9: THE GRAVITY EQUATION

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OVERVIEW

- **Readings:** Feenstra Chapter 5 and Chapter 6.
- Gravity equation: theory and evidence.

1. INTRODUCTION

- In the monopolistic competition model of trade, each country exports varieties of differentiated products to another: countries are completely specialized in different product varieties.
- In this case, trade flows can be described by the gravity equation.
- In its simplest form, gravity equation is $X = A\frac{Y_1 Y_2}{d^\rho}$.

2. BORDER EFFECTS IN THE GRAVITY MODEL

2-1. EVIDENCE

- Estimate the gravity equation using exports data between provinces and states in Canada and the U.S: both within and cross the border.
- Taking the logarithm on the gravity equation, adding indicator variables:
  $\ln X^{ij} = \beta_0 + \beta_1 \ln Y^i + \beta_2 \ln Y^j + \beta_3 \mathbf{1}_{\{i,j \in CAN\}} + \beta_4 \mathbf{1}_{\{i,j \in US\}} + \rho \ln d^{ij} + \varepsilon_{ij}.$
- Results:
  - $\beta_1 > 0$, $\beta_2 > 0$, and $\rho < 0$, not surprising.
  - $\beta_3 >> 0$ and $\beta_4 > 0$.

- β_3 is large, suggesting that cross-provincial trade (within Canada) is 15.6 (=e^{β_3}) times larger than cross-border trade (Canada-US): the **border effect**.
- $\beta_4 > 0$ but smaller: cross-state trade is 1.5 times larger than cross-border trade.
- Border effect is *asymmetric* on countries of different size: larger effect on small countries.
- Is the border effect exaggerated in estimation? Maybe, need a model to derive Gravity equation.

	McCallum (1995) and Other Samples			*Anderson and van Wincoop 2001*	*With Fixed Effects*[a]
	(1)	*(2)*	*(3)*	*(4)*	*(5)*
Year of data	*1988*	*1993*	*1993*	*1993*	*1993*
Regions included			US-US	US-US	US-US
	CA-CA	CA-CA	CA-CA	CA-CA	CA-CA
	CA-US	CA-US	CA-US	CA-US	CA-US
Independent variables					
$\ln Y^i$	1.21	1.22	1.13	1	1
	(0.03)	(0.04)	(0.02)		
$\ln Y^j$	1.06	0.98	0.97	1	1
	(0.03)	(0.03)	(0.02)		
$\ln d^{ij}$	−1.42	−1.35	−1.11	−0.79	−1.25
	(0.06)	(0.07)	(0.03)	(0.03)	(0.04)
Indicator Canada	3.09	2.80	2.75		
	(0.13)	(0.14)	(0.11)		
Indicator U.S.			0.40		
			(0.05)		
Indicator border				−1.65	−1.55
				(0.08)	(0.06)
Border effect Canada[b]	22.0	16.4	15.7	10.5	
	(2.9)	(2.0)	(1.9)	(1.2)	
Border effect U.S.[b]			1.5	2.6	
			(0.1)	(0.1)	
Border effect-average[c]			4.8	5.2	4.7
			(0.3)	(0.4)	(0.3)
R^2	0.81	0.76	0.85	n.a.	0.66
Observations	683	679	1511	1511	1511

2-2. MODEL

- In estimation, border indicator variables summarize all factors other than outputs and distances.
- What are the border effects?
  - Transport cost.
  - Tariffs.
  - ?
- We need to have a model to incorporate transport cost and tariffs: prices are no longer equalized cross border.
- Then the pattern of trade is more complex than in the gravity equation.
- We have seen that the iceberg trade cost leads to different prices of the same good in different markets.
- Price $p^{ij} = \tau^{ij} p^i$, export price from country i to country j is a τ_{ij} proportion of domestic price p^{ii}, and $\tau^{ij} \geq 1$.
  - Note the change of notation from Eaton-Kortum model.

- CES preference in country j: $U^j = \sum_{i=1}^{I} \sum_{k=1}^{N^i} (c_k^{ij})^{(\sigma-1)/\sigma}$.
  - c_k^{ij}, consumption of good k in country j imported from country i.
- N^i products, imposing symmetry between these products, so they all have the same price, $p_k^{ij} = p^{ij}$.
- Preference in country j becomes $U^j = \sum_{i=1}^{I} N^i (c^{ij})^{(\sigma-1)/\sigma}$.

- Budget constraint of the representative consumer in country j:
  $Y^j = \sum_{i=1}^{I} N^i p^{ij} c^{ij}$.

- Utility maximization in country j leads to
$$c^{ij} = \left(\frac{p^{ij}}{P^j}\right)^{-\sigma} \left(\frac{Y^j}{P^j}\right).$$
  - where the price index in country j is $P^j = [\sum_{i=1}^{I} N^i (p^{ij})^{1-\sigma}]^{1/(1-\sigma)}$.
- Total value of exports from country i to country j is, $X^{ij} = N^i p^{ij} c^{ij}$, then
$$X^{ij} = N^i Y^j \left(\frac{p^{ij}}{P^j}\right)^{1-\sigma}.$$

3. ESTIMATING THE GRAVITY EQUATION USING PRICE INDICES

- The number of products, N^i, is unobserved (not measured) in the data.
- Production: suppose that labor is the only input factor, and the production function is the same for all products (as we saw in the monopolistic competition model).
- Then GDP in country i is $Y^i = N^i p^i y$, or $N^i = Y^i/(p^i y)$. Recall $y = (\sigma - 1)\alpha\beta$ with CES preference and monopolistic competition.
- The value of exports from i to j becomes

$$X^{ij} = \frac{Y^i Y^j}{p^i y}\left(\frac{p^{ij}}{P^j}\right)^{1-\sigma} = \frac{Y^i Y^j}{(p^i)^\sigma y}\left(\frac{\tau^{ij}}{P^j}\right)^{1-\sigma}.$$

- Take logarithm and first differencing,

$$\Delta \ln X^{ij} = \Delta \ln(Y^i Y^j) + (1-\sigma)\Delta \ln \tau^{ij} - \sigma \Delta \ln p^i + (\sigma - 1)\Delta \ln P^j.$$

  - Note p^i is the price of country i's exported goods but also sold in country i. P^j is the overall price level in country j.
  - τ^{ij} includes tariffs and transport costs: observed (measured) in the data.
- Further let $s^i = \frac{Y^i}{Y^i + Y^j}$, then $Y^i Y^j = (Y^i + Y^j)^2 s^i s^j$. We have

$$\Delta \ln X^{ij} = 2\Delta \ln(Y^i + Y^j) + \Delta \ln(s^i s^j) + (1-\sigma)\Delta \ln \tau^{ij} - \sigma \Delta \ln p^i + (\sigma - 1)\Delta \ln P^j.$$

  - $s^i s^j = [1 - (s^i)^2 - (s^j)^2]/2$, it measures the "size dispersion" of countries: $s^i s^j$ becomes *larger* as countries become more *similar*.

- Estimation, Baier and Bergstrand (2001).
- Data: 16 OECD countries, difference between two periods: 1958-1960 and 1986-1988.
- Results: $R^2 = .4$; $N = 240$.

$$\Delta \ln X^{ij} = .05 + 2.4\Delta \ln(Y^i + Y^j) + .6\Delta \ln(s^i s^j) - 4.5\Delta \ln(1 + t^{ij}) - 3.2\Delta \ln (1 + a^{ij}) - .7\Delta \ln Y^j - .25\Delta \ln(p^i / P^j) - .1 \ln X_0^{ij}.$$

  - t^{ij} is tariff, a^{ij} is transport cost: both are statistically significant.
  - Positive estimate for $s^i s^j$: countries of similar sizes trade more.
  - Extra term $\ln Y^j$, due to a more general model than we presented here.
  - $R^2 = .4$: the gravity equations explains close to a half of the change in bilateral trade flows in OECD countries.
- Sum of GDP, tariffs, and transport costs explain the *entire increase* in trade, leaving no role for $s^i s^j$.

4. ESTIMATING GRAVITY EQUATION USING ESTIMATED BORDER EFFECTS

- Anderson and van Wincoop (2003).
- Relative prices may be inadequate to measure the trade frictions.
  - Other factors than tariffs and transport costs.
- We can specify: $\ln \tau^{ij} = \delta^{ij} + \rho \ln d^{ij} + \varepsilon_{ij}$.
  - δ^{ij} can be indicator variable for border effects (in addition to distance).
- Substitute this expression for τ^{ij} in $X^{ij} = \frac{Y^i Y^j}{(p^i)^\sigma y} \left(\frac{\tau^{ij}}{P^j} \right)^{1-\sigma}$. The resulted equation is highly nonlinear, difficult to estimate.
- Need to exploit the model further.

- Total value of sales of good i is $p^i y^i = \sum_{j=1}^{I} p^{ij} c^{ij}$, then $y^i = \sum_{j=1}^{I} c^{ij} \tau^{ij}$.
- Theorem due to Anderson and van Wincoop (2003)

> *Suppose that the transportation costs are symmetric, i.e., $\tau^{ij} = \tau^{ji}$. An implicit solution to the market clearing condition $y^i = \sum_{j=1}^{I} c^{ij} \tau^{ij}$ is $\tilde{p^i} \equiv (s^i / N^i)^{1/(1-\sigma)} / \tilde{P^i}$. The price indices are solved by $(\tilde{P^j})^{1-\sigma} = \sum_{i=1}^{I} s^i (\tau^{ij} / \tilde{P^i})^{1-\sigma}$.*

- Here, $s^i = Y^i / \sum_{j=1}^{I} Y^j$.

- **Proof**: First recall that $c^{ij} = \left(\frac{p^{ij}}{P^j}\right)^{-\sigma}\left(\frac{Y^j}{P^j}\right)$. Then GDP in country i is $Y^i = N^i p^i y^i = N^i \sum_{j=1}^I c^{ij} p^{ij} = N^i \sum_{j=1}^I (\tau^{ij} p^i / P^j)^{1-\sigma} Y^j$. Now replace p^i with $\tilde{p}^i$, we have

$$Y^i = s^i \sum_{j=1}^I \left(\frac{\tau^{ij}}{\tilde{P}^i \tilde{P}^j}\right)^{1-\sigma} Y^j.$$

Adding Y^i for all i, we have $Y^w = \sum_{i=1}^I Y^i = \sum_{j=1}^I \left(\frac{\tau^{ij}}{\tilde{P}^i \tilde{P}^j}\right)^{1-\sigma} Y^j$. Divide both sides by Y^w and multiply both sides by $(\tilde{P}^i)^{1-\sigma}$, we obtain

$$(\tilde{P}^i)^{1-\sigma} = \sum_{j=1}^I s^j \left(\frac{\tau^{ij}}{\tilde{P}^j}\right)^{1-\sigma}.$$

On the other hand, we can use the price index obtained earlier, $P^i = [\sum_{j=1}^I N^j (p^{ij})^{1-\sigma}]^{1/(1-\sigma)}$. Replace $p^{ij} = p^i \tau^{ij}$ and set $p^i = \tilde{p}^i$, we can also obtain the expression for $\tilde{P}^i$. ■

- Recall the exports from i to j was $X^{ij} = N^i Y^j \left(\frac{p^{ij}}{P^j}\right)^{1-\sigma}$. Use the implicit price $\tilde{p}^i$ and the implicit price index $\tilde{P}^i$, we have

$$X^{ij} = s^i Y^j \left(\frac{\tau^{ij}}{\tilde{P}^i \tilde{P}^j}\right)^{1-\sigma},$$

or

$$X^{ij} = \frac{Y^i Y^j}{Y^w} \left(\frac{\tau^{ij}}{\tilde{P}^i \tilde{P}^j}\right)^{1-\sigma}.$$

- $\tilde{P}^i$ is called "indices of multilateral resistance" because they depend on transportation costs τ^{ij}.

- Take the logarithm of the exports X^{ij}, omitting the constant term Y^w.

$$\ln(X^{ij}/Y^iY^j) = (1-\sigma)\ln\tau^{ij} + \ln(\tilde{P}^i\tilde{P}^j)^{\sigma-1}.$$

- Now substitute $\ln\tau^{ij} = \delta^{ij} + \rho\ln d^{ij} + \varepsilon_{ij}$ for $\ln\tau^{ij}$, we have

$$\ln(X^{ij}/Y^iY^j) = \rho(1-\sigma)\ln d^{ij} + (1-\sigma)\delta^{ij} + \ln(\tilde{P}^i\tilde{P}^j)^{\sigma-1} + (1-\sigma)\varepsilon_{ij}.$$

- The estimation equation can be written as

$$\ln(X^{ij}/Y^iY^j) = \alpha \ln d^{ij} + \gamma\delta^{ij} + \ln(\tilde{P}^i\tilde{P}^j)^{\sigma-1} + (1-\sigma)\varepsilon_{ij}.$$

  - $\delta^{ij} = 1$ if X^{ij} is between Canada and the U.S., =0 otherwise.

- **Estimation**: iterate on the two steps as follows

  - Step 1: Minimize the sum of squared residuals in the above equation, and simultaneously obtain τ^{ij} within the minimization.

  - Step 2: Solve for $(\tilde{P}^i)^{\sigma-1}$ from the Theorem.

- Data: Same data as in the above table.

- Estimates are shown as column (4) in the above table: $\widehat{\alpha} = -.79, \widehat{\gamma} = -1.65.$

- $\widehat{\gamma} = -1.65$, border effect is $e^{1.65} = 5.2$. See the Feenstra page 142 for interpretations.

Extension

- The above estimation requires some programming, cannot be done with ordinary least squares (OLS).
- An alternative estimation is to use the **fixed effects**
- Let $\delta_1^i = 1$ if region i is the exporter in trade X^{ij}, =0 otherwise;
- Let $\delta_2^j = 1$ if region j is the importer, =0 otherwise.
- We can run an OLS as follows
$$\ln(X^{ij}/Y^iY^j) = \alpha \ln d^{ij} + \gamma\delta^{ij} + \beta_1^i\delta_1^i + \beta_2^j\delta_2^j + (1-\sigma)\varepsilon_{ij}.$$
- Estimates: $\widehat{\gamma} = 1.55$. The border effect is $e^1.55 = 4.7$. Column (5) in the above table.